

FUNCTIONS AND GRAPHS

Directions for questions 1 to 6: Select the correct alternative from the given choices.

The graph $y = x^2$ is symmetric about ____.

- ☐ a) y - axis
- ☐ b) x - axis
- ☐ c) $x = 1$
- ☐ d) $y = 1$

Solution:

Let $f(x) = x^2$
 $f(-x) = (-x)^2 = x^2$
 $\therefore f(-x) = f(x)$
 $\therefore$ The graph is symmetric about y - axis.

Choice (A)

Directions for questions 1 to 6: Select the correct alternative from the given choices.

The reflection of the point $(-5, -7)$ in the x - axis is ____.

☐ a) $(5, 7)$

☒ b) $(-5, 7)$ (Your answer - ✓)

☐ c) $(5, -7)$

☐ d) $(-5, -7)$

Directions for questions 1 to 6: Select the correct alternative from the given choices.

The graph $y = \log x$ passes through _____.

☐ a) Q_3, Q_4

☒ b) Q_1, Q_2 (Your answer - ✖)

☐ c) Q_2, Q_3

☐ d) Q_1, Q_4

Solution:

Since $\log x$ is only defined for positive real numbers, $x > 0$. $y = \log x$ can be positive or negative, so the graph $y = \log x$ lies in Q_1 and Q_4 only. Choice (D)

Directions for questions 1 to 6: Select the correct alternative from the given choices.

If $y = f(x)$ is a function, the graph of $y = f(x - 2)$ is obtained by shifting $f(x)$ _____.

☐ a) towards left by 2 units.

☐ b) vertically upwards by 2 units.

☐ c) towards right by 2 units.

☐ d) vertically downwards by 2 units.

Solution:

When $f(x)$ is moving 2 units right side, the transformed function is $f(x - 2)$. Choice (C)

Directions for questions 1 to 6: Select the correct alternative from the given choices.

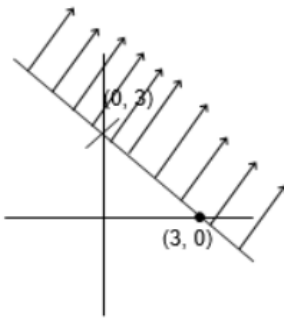
The reflection of $y = f(x)$ in the y - axis is _____.

- ☐ a) $-f(x)$
- ☐ b) $f(-x)$
- ☐ c) $-f(-x)$
- ☐ d) None of these

Solution:

The reflection of $f(x)$ w. r. t to y axis is $f(-x)$.
(Standard result). Choice (B)

The inequality that represents the region shown in the graph is _____.



- ☐ a) $x - y \geq 3$
- ☐ b) $x + y \leq 3$
- ☐ c) $x + y \geq 3$
- ☐ d) $x - y \leq 3$

Solution:

The equation of the line passing through (3, 0) and (0, 3) is $x + y = 3$.
Origin is not lying in the region, so required in equation is $x + y \geq 3$.

Choice (C)

Directions for questions 7 to 10: Type in your answer in the input box provided in the question.

A, B are finite sets and $n(A) = 3$, $n(B) = 5$. The number of functions defined from B to A is

Your Answer: 125 ✖

Solution:

Given $n(B) = 5$; $n(A) = 3$

The number of functions defined from B to A is $[n(A)]^{n(B)} = 3^5 = 243$.

Ans : (243)

Directions for questions 7 to 10: Type in your answer in the input box provided in the question.

If $f(x) + f(1 - x) = 2$, and $x \in \mathbb{R}$;

$$f\left(\frac{1}{3}\right) + f\left(\frac{2}{3}\right)$$

=

Solution:

Given $f(x) + f(1 - x) = 2$

Put $x = \frac{1}{3}$; $f\left(\frac{1}{3}\right) + \left(1 - \frac{1}{3}\right) = 2$

$$f\left(\frac{1}{3}\right) + f\left(\frac{2}{3}\right) = 2$$

Ans : (2)

Directions for questions 7 to 10: Type in your answer in the input box provided in the question.

$x \in \mathbb{R}$ and $f(x) = x^2 + 3$; $g(x) = 2x + 5$. $\text{fog}(4) =$

Solution:

$$\begin{aligned}\text{Given } f(x) &= x^2 + 3; g(x) = 2x + 5 \\ \text{fog}(4) &= f[g(4)] \\ &= f[13] = (13)^2 + 3 \\ &= 172\end{aligned}$$

Ans : (172)

Directions for questions 7 to 10: Type in your answer in the input box provided in the question.

If $f(x) = 2x + 3$, $x \in \mathbb{R}$ then $f^{-1}(3) =$

Solution:

$$\begin{aligned}\text{Given } f(x) &= 2x + 3 \\ \text{Let } f^{-1}(3) &= k \\ \therefore f(k) &= 3 \\ \Rightarrow 2k + 3 &= 3 \\ \Rightarrow k &= 0 \\ \Rightarrow f^{-1}(3) &= 0\end{aligned}$$

Ans : (0)

Directions for questions 11 to 13: Select the correct alternative from the given choices.

The range of $|x + 5|$ is _____.

- ☐ a) $[5, \infty)$
- ☐ b) $(0, \infty)$
- ☐ c) $[0, \infty)$
- ☐ d) $(5, \infty)$

Solution:

The range of $|x + 5|$ is $[0, \infty)$, as modulus of any quantity is non-negative.

Choice (C)

Directions for questions 11 to 13: Select the correct alternative from the given choices.

The domain of the function $\log(x - 3)$ is _____.

- ☐ a) $(3, \infty)$
- ☐ b) $[3, \infty)$
- ☐ c) $(0, \infty)$
- ☐ d) $(-\infty, \infty)$

Solution:

We know $\log f(x)$ is defined only when $f(x) > 0$

$\therefore \log(x - 3)$ is defined when

$$x - 3 > 0 \Rightarrow x > 3$$

$\therefore$ domain of the function is $(3, \infty)$

Choice (A)

Directions for questions 11 to 13: Select the correct alternative from the given choices.

If the range of a function is equal to its co-domain, then the function is called ____.

- ☐ a) one – to- one function
- ☐ b) bijective function
- ☐ c) onto function
- ☐ d) constant function

Solution:

Onto function (By definition)

Choice (C)

Directions for question 14: Type in your answer in the input box provided in the question.

The number of one-to-one functions defined from A to B, where $n(A) = 4$ and $n(B) = 2$ is

Solution:

If $n(A) > n(B)$, then the number of one-to-one functions is 0.

Ans : (0)

Directions for question 15: Select the correct alternative from the given choices.

$f : A$

$\rightarrow$

B is a function. If the relation $f^{-1} : B$

$\rightarrow$

A is also a function, which of the following is true?

- ☐ a) $f : A \rightarrow B$ is only a one-to-one function
- ☐ b) $f : A \rightarrow B$ is only an onto function
- ☐ c) $f : A \rightarrow B$ is a bijective function
- ☐ d) None of these

Solution:

If $f^{-1} : B \rightarrow A$ is a function, then
 $f : A \rightarrow B$ must be a bijective function.

Choice (C)